Canto

Displaced-track reconstruction, trained on ttbar only

	Canto-L 3 seeds · full budget	Canto-H 3 seeds · full budget	Canto-B 3 seeds · full budget	acorn (HNL) 3 seeds · HNL-trained	acorn 3 seeds · ttbar-trained	DeZoort IN 1 seed · ttbar-trained
Displaced tracks |d0| ≥ 10 mm or r_v ≥ 100 mm	**99.1%**	**99.1%**	98.4%	95.6%	92.3%	84.8%
Far-displaced tracks |d0| 100–300 mm	**97.6%**	96.6%	96.2%	94.4%	89.2%	80.2%
Very far-displaced |d0| ≥ 300 mm	**87.7%** 50/57 particles	82.5% 47/57 particles	78.9% 45/57 particles	80.7% 46/57 particles	64.9% 37/57 particles	47.4% 9/19 particles
Prompt muons W → μ in HNL events	**99.8%**	**99.8%**	99.7%	98.8%	98.7%	91.2%
Dense prompt tracks ttbar, |d0| < 1 mm	96.3%	96.2%	**96.6%**	84.6%	87.2%	62.6%
Fake-track rate lower is better	**1.03%**	1.43%	1.30%	2.92%	2.97%	12.51%
Training data	ttbar only	ttbar only	ttbar only	HNL signal	ttbar only	ttbar only
Parameters edge classifier	348k	348k	348k	264k	264k	6.4k

All numbers: ACTS generic detector simulation without pileup, validation set (2,999 events of ttbar and the two HNL samples: the displaced-track, prompt-muon and fake-rate rows from the HNL events, 'Dense prompt tracks' from the ttbar events); every model at a track fake rate ≤ 2.94 % (pooled HNL val, the preregistered working point FR*) where it can reach it, otherwise at its lowest fake rate (acorn ttbar-trained seeds 0 and 2, acorn HNL-trained seed 1, DeZoort IN); track efficiency by double-majority matching; mean over the listed training seeds. Best value per row bold and shaded (ties included).

Canto (Chained-triplet, origin-Agnostic Networks for Tracking Off-vertex), trained on ttbar only; the branches differ in the graph: Canto-B the base graph, Canto-H a hand-cut helix-triplet union, Canto-L a learned triplet-support union. All three at the full budget (30 epochs on the ttbar training graphs), three seeds each; the builder of each branch chosen on val by rule R6 with the one-standard-error rule (within one binomial standard error of the best efficiency, the lowest fake rate): Canto-L learned merging, Canto-H walkthrough + merging, Canto-B learned fragment merging. The one-time test evaluation is pending.

acorn (HNL): acorn's interaction network with connected components, trained on the HNL signal as in Wang et al. (arXiv:2203.08800), whose own pipeline used an attention GNN and DBSCAN; it reproduces their all-track efficiency (test: 0.9554 vs 0.9561 in the paper). acorn: the same network trained on ttbar only, like Canto. Both at their highest-efficiency cut within the fake-rate target where reachable, the most favourable choice for acorn (with the preregistered cut rule: 93.8 % / 92.3 %).

DeZoort IN: the original 6.4k-parameter interaction network (DeZoort et al. 2021), trained on ttbar here; it cannot reach the fake-rate target and is shown at its lowest fake rate (12.5 %). |d0| ≥ 300 mm: 19 particles per seed, statistics limited.